



24BCF10059 - Lakshdeep

Experiment 5

Aim :

To study and implement advanced SQL techniques such as conditional/custom ordering using CASE expressions inside ORDER BY, and self-joins for modelling hierarchical (manager–employee) relationships.

Question 1

A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion. For each registered user, calculate the confirmation rate, which is defined as:

$$\text{Confirmation Rate} = \frac{\text{Number of Successful Confirmations}}{\text{Total Confirmation Requests}}$$

Solution :

```
SELECT U.user_id ,  
ROUND(  
    AVG(  
        CASE  
            WHEN C.action = 'confirmed' THEN 1  
            WHEN C.action = 'timeout' THEN 0  
            ELSE 0  
        END  
    ),2  
) AS Confirmation_Rate  
FROM Signups as U  
LEFT JOIN  
Confirmations as C  
ON U.user_id = C.user_id  
GROUP BY U.user_id
```

Question 2

A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between **2013-10-01** and **2013-10-03**, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

Solution :

```
SELECT T.request_at,ROUND(
    AVG(
        CASE
            WHEN T.status = 'cancelled_by_driver' THEN 1
            WHEN T.status = 'cancelled_by_client' THEN 1
            ELSE 0
        END
    ),2
) AS "Cancellation_rate_for_not_banned_clients"
FROM Users AS C
JOIN
Trips AS T
ON C.users_id = T.client_id
JOIN
Users AS D
ON D.users_id = T.driver_id
WHERE C.banned = 'No'
AND D.banned = 'No'
AND T.request_at BETWEEN '2013-10-01' AND '2013-10-03'
GROUP BY T.request_at;
```

Learning Outcomes :

After completing this experiment, the following learning outcomes were achieved.

- Understood how to use **LEFT JOIN** to include all registered users, even when they have no confirmation records.
- Learned to calculate **confirmation rates using AVG() and CASE expressions** based on different confirmation actions.
- Applied **ROUND()** to present calculated rates with a specified number of decimal places.
- Strengthened understanding of **GROUP BY** for generating user-wise summary results.
- Learned to identify valid trips by applying conditions on **both customers and drivers** using JOIN operations.
- Applied **WHERE conditions and date filtering** to analyze trip records within a specified date range.
- Understood how to calculate **cancellation rates using conditional aggregation**, such as counting canceled trips relative to total valid trips.